

LOCAL REPAIRED-GAUGE REPRESENTATION FOR SINGLE-CORE TYPE II NAVIER-STOKES CONCENTRATION

ABSTRACT. We prove the local representation theorem needed to place a genuine single-core Type II Navier–Stokes concentration into the associated compactness criterion. Starting from a positive local CKN core with compact-cylinder suitable bounds and a unique nondegenerate scale and center, we construct a local renormalized chart, compactly supported repaired gauges, local pressure representation, and modulation coefficients. The resulting repaired orbit satisfies the local single-core compactness criterion. The pressure reconstruction and the modulation coefficients are consequences of the local chart and repaired gauge; they are not additional assumptions. Failure of the localized gauge Jacobian or of compact-cylinder uniqueness places the sequence in the multibubble or cascade configurations. This paper uses the classical Navier–Stokes theory for suitable weak solutions (Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], and Lin [Lin98]).

1. SINGLE-CORE INPUT

Let (u, p) be a suitable weak solution near a possible singular point (x_0, T) . A local Type II concentration sequence is a sequence $r_n \downarrow 0$ such that the parabolic rescalings

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s)$$

are suitable on every fixed backward cylinder and retain positive local CKN mass on one fixed cylinder.

For a suitable pair (v, q) , let $A_v, E_v, C_{v,q}, D_{v,q}$ denote the usual scale-invariant local energy, dissipation, velocity, and pressure quantities on backward cylinders. The single-core case is the case in which:

- (i) $A + E + C + D$ is bounded on every compact rescaled cylinder;
- (ii) one selected core carries a fixed positive CKN density;
- (iii) outside a larger compact cylinder, no comparable active core remains in the local rescaled frame;
- (iv) the localized scale and center selected by that core are unique in the sense that the localized scale-center Jacobian is invertible.

If any of these conditions fails, the case is not single-core and belongs to the multibubble/cascade analysis.

2. LOCAL CHART AND EQUATION

Choose absolutely continuous local scale and center functions $\Lambda(\tau) > 0$ and $X(\tau)$, and set

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \quad \frac{d\tau}{dt} = \Lambda(\tau)^{-2}, \quad u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau).$$

On compact cylinders, the transformed velocity satisfies

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

where

$$a(\tau) = \partial_\tau \log \Lambda(\tau), \quad b(\tau) = -\Lambda(\tau)^{-1} \partial_\tau X(\tau)$$

up to the harmless sign convention determined by the chart. Only consistency with the displayed equation is used below.

Lemma 2.1 (Local change of variables). *Let $\Lambda > 0$ and X be absolutely continuous on a compact time window, and define V, P, a, b by the chart above. Then the transformed fields solve the displayed renormalized Navier–Stokes equation in the sense of distributions on compact cylinders.*

Proof. This is the standard distributional change of variables for Navier–Stokes under an absolutely continuous translation and dilation. The terms containing $\partial_\tau \log \Lambda$ and $\partial_\tau X$ are exactly the scale and translation modulation terms in the displayed equation. $\square$

3. LOCALIZED REPAIRED GAUGES

Fix R and $\chi_R \in C_c^\infty(B_{2R})$, $\chi_R = 1$ on B_R . Set

$$G_{0,R}(V) = \int \chi_R(Y) |Y|^{-p} |V(Y)|^3 dY - \Theta_0, \quad 0 < p < 3,$$

$$G_{j,R}(V) = \int Y_j \chi_R(Y) |V(Y)|^2 dY, \quad j = 1, 2, 3.$$

Lemma 3.1 (Scale row). *For variations generated by scaling,*

$$DG_{0,R}(V)[V + Y \cdot \nabla V] = p \int \chi_R |Y|^{-p} |V|^3 - \int (Y \cdot \nabla \chi_R) |Y|^{-p} |V|^3.$$

Thus, if the cutoff-annulus term is strictly smaller than the weighted core contribution, the scale row is transverse.

Proof. Differentiate $G_{0,R}(\lambda V(\lambda \cdot))$ at $\lambda = 1$. The homogeneity of $|Y|^{-p}$ gives the first term, and differentiating the cutoff gives the annular error. $\square$

Theorem 3.2 (Local repaired-gauge solve). *On every compact time window where the localized scale-center Jacobian is invertible with bounded inverse, there are absolutely continuous local scale and center corrections imposing the four constraints. If this localized Jacobian is not invertible, the branch is not a nondegenerate single-core branch and belongs to the multibubble, cascade, or gauge-degenerate alternative.*

Proof. The constraints are continuously differentiable with respect to the finite number of local scale and center parameters on the compact support of χ_R . The quantitative inverse function theorem gives a unique correction while the Jacobian remains invertible. Absolute continuity follows from the absolute continuity of the unrepaired chart and the local integrability of the time derivatives of the constraints. Failure of the Jacobian means that the selected mass does not determine one local scale and center, which is precisely the non-single-core alternative. $\square$

4. PRESSURE AND MODULATION

For every compact ball, decompose

$$P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j),$$

where $\zeta \in C_c^\infty(B_{2R})$ equals one on the smaller ball. Then H is harmonic on the smaller ball. Calderon–Zygmund estimates for P_{loc} and interior harmonic estimates for H give the pressure bounds used in local CKN and Caccioppoli estimates.

Proposition 4.1 (Modulation coefficients). *On every compact selected window on which the localized gauge Jacobian remains invertible and the local Caccioppoli bounds hold, the repaired chart has measurable modulation coefficients a, b that are locally bounded in terms of the inverse Jacobian, the local pressure bounds, and the compact-cylinder suitable norms.*

Proof. Differentiate the four gauge constraints in time. The coefficients a, b appear linearly through the scale and translation rows of the renormalized equation. The coefficient matrix is the localized gauge Jacobian. The forcing terms consist of local convection, viscosity, and pressure contributions, all estimated on the compact support of χ_R by Caccioppoli and the local pressure decomposition. Inverting the Jacobian gives the asserted local bounds. $\square$

5. REPRESENTATION THEOREM

Theorem 5.1 (Single-core repaired-gauge representation). *Let a local Type II concentration sequence be single-core in the sense of Section 1. On every compact window on which the selected core determines an absolutely continuous local rescaled frame and the localized scale-center Jacobian is invertible, after passing to a subsequence the branch admits a localized repaired-gauge orbit (V, P, a, b) satisfying:*

- (i) *the renormalized Navier–Stokes equation on compact cylinders;*
- (ii) *positive selected local CKN mass on the selected core;*
- (iii) *local pressure reconstruction modulo spatial means;*
- (iv) *locally bounded modulation coefficients on the selected windows;*
- (v) *the compact-cylinder suitable and Caccioppoli bounds used by the local single-core compactness criterion.*

The pressure reconstruction and the coefficients a, b in items (iii)–(iv) are conclusions of the theorem. The only structural alternatives to the conclusion are absence of a single absolutely continuous local frame, failure of localized gauge nondegeneracy, or loss of compact-cylinder suitable bounds; these are precisely the multibubble, cascade, gauge-degenerate, or local compactness-failure cases.

Proof. The local rescaled frame and [Theorem 2.1](#) give the renormalized equation before repair. The repaired-gauge solve imposes the localized scale and centering constraints and preserves absolute continuity of the corrected parameters. The pressure reconstruction follows from the Navier–Stokes pressure equation under the same change of variables, modulo functions of time, and the compact-ball decomposition gives the local pressure bounds. The coefficients a, b are forced by the absolutely continuous scale and translation parameters in the transformed equation; [Theorem 4.1](#) gives their local bounds on selected windows. The selected CKN mass and compact-cylinder suitable bounds define the nondegenerate compact single-core branch. Thus the resulting orbit satisfies the local bounds used by the local single-core compactness criterion. Failure of the localized scale/center uniqueness is the non-single-core alternative. $\square$

REFERENCES

- [CKN82] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. Partial regularity of suitable weak solutions of the navier–stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.
- [Ler34] Jean Leray. Sur le mouvement d’un liquide visqueux emplissant l’espace. *Acta Mathematica*, 63:193–248, 1934.
- [Lin98] Fanghua Lin. A new proof of the caffarelli-kohn-nirenberg theorem. *Communications on Pure and Applied Mathematics*, 51(3):241–257, 1998.
- [Sch76] Vladimir Scheffer. Partial regularity of solutions to the navier-stokes equations. *Pacific Journal of Mathematics*, 66(2):535–552, 1976.